Double Random Phase Encoded Image Decryption Based on Deep Neural Networks

Abstract

Double Random Phase Encoding (DRPE) is a classical optical encryption technique widely regarded as secure due to the secrecy of its phase keys. However, the growing capability of deep learning models has raised concerns about its security against data-driven attacks. In this paper, we investigate the feasibility of reconstructing plaintext images from DRPE-encrypted magnitude-only representations using deep neural networks.

A UNet-based reconstruction framework with attention mechanisms is employed to learn the inverse mapping from encrypted data to original images. Different loss function strategies are evaluated to analyze their impact on reconstruction quality. Experimental results show that attention-enhanced networks trained with pixel-wise loss achieve stable convergence and superior visual fidelity, while perceptual and multi-loss refinements provide only limited improvements. This behavior indicates that magnitude-only encryption introduces irreversible information loss that fundamentally constrains reconstruction accuracy.

These findings demonstrate that meaningful semantic information can be recovered without access to encryption keys, suggesting that the security of traditional DRPE systems should be re-examined under modern learning-based threat models.

Keywords: Double random phase encryption; Neural network; Irreversible encryption system;

1 Introduction

In recent years, with the rapid development of digital technologies, the security of optical information processing systems becomes a hot spot, drawing significant attention to the field of image encryption. Among a variety of optical encryption methods, the double random phase encoding (DRPE) scheme, which was proposed by Refregier and Javidi in 1995, has long been regarded as an unbreakable optical encryption technique, owing to its reliance on two statistically independent random phase masks in the input and Fourier planes [1].

However, the advent of machine learning, particularly deep neural networks, has brought some possible changes to the field. Data-driven models have demonstrated an unprecedented ability to approximate complicated mappings, including those previously considered analytically or computationally intractable. This progress has raised the question of whether DRPE and related optical encryption systems remain secure in the presence of modern neural-network-based reconstruction techniques.

In this work, we present a learning-based framework capable of reconstructing plaintext images from DRPE-encrypted ciphertexts under realistic imaging conditions. We design and train a U-Net–based architecture, enhanced by multiple perceptual and structural loss terms, to recover high-fidelity images. Our results show that deep neural networks can reveal spatial structures that were previously inaccessible through traditional numerical decryption, thereby providing new insights into the vulnerabilities of classical optical encryption schemes.

2 Background and Methodology

2.1 Principle of Double Random Phase Encoding

Double Random Phase Encoding (DRPE) is a classical optical image encryption scheme originally proposed by Refregier and Javidi [1]. By exploiting the properties of Fourier transformation and random phase modulation, DRPE converts a meaningful image into a noise-like complex distribution, making direct visual interpretation infeasible.

Let $I(x, y)$ denote a grayscale input image of size $M \times N$. Two statistically independent random phase masks are generated: $P_1(x, y)$ in the spatial domain and $P_2(u, v)$ in the frequency domain, defined as

$$P_1(x, y) = \exp[j2\pi\phi_1(x, y)], \tag{1}$$

$$P_2(u, v) = \exp[j2\pi\phi_2(u, v)], \tag{2}$$

where $\phi_1(x, y)$ and $\phi_2(u, v)$ are independent random variables uniformly distributed in $[0, 1)$, and j denotes the imaginary unit.

The DRPE encryption process consists of the following steps:

1. The input image is first multiplied by the spatial-domain random phase mask:

$$I_1(x, y) = I(x, y) \cdot P_1(x, y). \tag{3}$$

2. A Fourier transform is applied to obtain the frequency-domain representation:

$$F_1(u, v) = \mathcal{F}\{I_1(x, y)\}. \tag{4}$$

3. The resulting spectrum is modulated by the frequency-domain random phase mask:

$$F_2(u, v) = F_1(u, v) \cdot P_2(u, v). \tag{5}$$

4. Finally, an inverse Fourier transform is performed to generate the encrypted image:

$$E(x, y) = \mathcal{F}^{-1}\{F_2(u, v)\}. \tag{6}$$

The encrypted output $E(x, y)$ is a complex-valued image with a noise-like appearance. In practical implementations, its real and imaginary parts, or alternatively its magnitude and phase components, are used as the encrypted data for transmission or further processing.

Figure 1 illustrates the overall workflow of the DRPE encryption process, including spatial-domain phase modulation, Fourier-domain encoding, and inverse transformation.

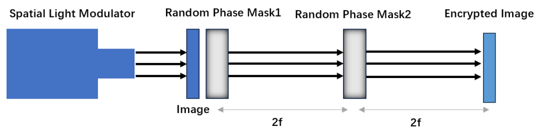

Figure 1: Schematic diagram of the Double Random Phase Encoding (DRPE) encryption process.

2.2 U-Net-based Decryption Network

In this work, to recover plaintext images from DRPE-encrypted image, a deep convolutional neural network is employed to learn an approximate inverse mapping between encrypted images and their corresponding original images. Among various architectures, the U-Net has demonstrated strong performance in image-to-image translation tasks due to its symmetric encoder-decoder structure and multi-scale feature fusion capability.

In this work, the encrypted image serves as the network input, while the corresponding plaintext image is used as the supervision target. The network is trained in a fully supervised manner to minimize the discrepancy between the reconstructed output and the ground truth image.

2.2.1 Standard U-Net Architecture

The standard U-Net architecture was first introduced by Ronneberger [2] for biomedical image segmentation. Its structure consists of a contracting path (encoder) and an expansive path (decoder), connected by skip connections at corresponding resolutions.

The encoder is used to extract features through repeated convolution and downsampling operations, which could get the abstract representations of the input image. Conversely, the decoder performs upsampling and convolution to reconstruct spatial content and details. The skip connections concatenate feature maps from the encoder to the decoder, preserving fine-grained spatial information during reconstruction.

Such a design is particularly suitable for DRPE decryption tasks, as it enables the network to jointly utilize global contextual information and local structural details, both of which are essential for recovering meaningful content from noise-like encrypted images.

Figure 2 shows the schematic structure of the standard U-Net architecture.

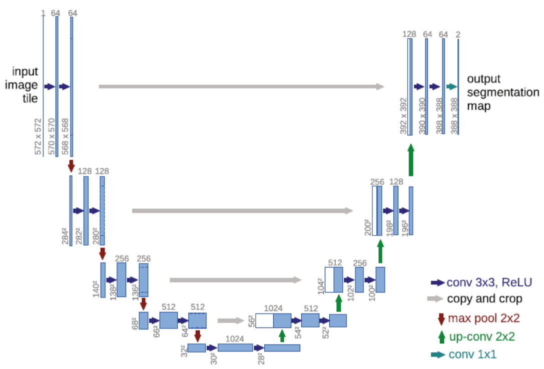

Figure 2: Structure of a standard U-Net architecture [2].

2.2.2 U-Net with Attention Mechanism

Although the standard U-Net effectively combines multi-scale features, it treats all spatial regions and feature channels with equal importance. However, in DRPE decryption, for encrypted images, useful information is often sparsely distributed within complex noise-like patterns; for decrypted image, we don't care about the background of the unencrypted image, instead, we focus on the main subject but the model could avoid punishment by just recovering the simple and easily recoverable background.

To address this issue, an attention mechanism is incorporated into the U-Net framework. Attention modules allow the network to adaptively emphasize informative regions and suppress irrelevant or noisy features, thereby improving reconstruction quality[3]. In this work, attention gates are integrated into the skip connections, enabling the decoder to selectively receive encoder features based on their relevance to the current decoding stage.

The concept of attention-enhanced U-Net architectures has been explored in medical image analysis and related image reconstruction tasks, where attention gates are shown to improve focus on task-relevant structures[4]. The overall design in this study follows the general U-Net framework proposed by Ronneberger, with additional attention-based feature modulation to enhance decryption performance.

By introducing attention mechanisms, the network gains improved capability to distinguish meaningful signal components from encryption-induced noise, which is particularly beneficial when handling complex DRPE-encrypted images.

2.3 Loss Functions and Training Objective

The decryption of DRPE-encrypted images is formulated as a supervised image reconstruction problem. Given an encrypted image as input, the network is trained to minimize the difference between the reconstructed output and the corresponding plaintext image. To achieve stable convergence and high-quality reconstruction, multiple loss functions are investigated in this work.

2.3.1 L1 Loss

The L1 loss measures the average absolute difference between the reconstructed image $\hat{I}$ and the ground truth image I:

$$\mathcal{L}_{\mathrm{L1}} = \frac{1}{N}\sum_{i=1}^{N}\left|\hat{I}_i - I_i\right|, \tag{7}$$

where N denotes the total number of pixels.

Compared to the L2 loss(Mean Square loss), the L1 loss is less sensitive to outliers and encourages sharper reconstructions. In the context of DRPE decryption, L1 loss provides strong pixel-wise supervision and serves as a stable baseline for network training.

2.3.2 Structural Similarity (SSIM) Loss

While pixel-wise losses focus on absolute intensity differences, they often fail to capture perceptual image quality. To address this limitation, the Structural Similarity Index Measure (SSIM) is adopted to evaluate structural consistency between images.

SSIM compares two images based on luminance, contrast, and structural information, and is defined as:

$$\mathrm{SSIM}(I, \hat{I}) = \frac{(2\mu_I \mu_{\hat{I}} + C_1)(2\sigma_{I\hat{I}} + C_2)}{(\mu_I^2 + \mu_{\hat{I}}^2 + C_1)(\sigma_I^2 + \sigma_{\hat{I}}^2 + C_2)}, \tag{8}$$

where μ, σ^2, and $\sigma_{I\hat{I}}$ denote the mean, variance, and covariance, respectively.

In this work, SSIM is converted into a loss function as:

$$\mathcal{L}_{\mathrm{SSIM}} = 1 - \mathrm{SSIM}(I, \hat{I}). \tag{9}$$

By emphasizing structural similarity, SSIM loss encourages the preservation of image contours and global structures, which are crucial for visually meaningful decryption results.

However, due to SSIM's emphasis on structural similarity and neglect of texture details, it often falls into the dilemma of extremely high contrast and approaching binary values[5]. As a result, in this work, SSIM is used as a plus to the L1 loss.

2.3.3 Additional Regularization Losses

To further enhance reconstruction quality, especially the details, additional loss terms are incorporated, including perceptual loss, edge-aware loss, total variation (TV) loss, and Dice-based edge loss.

The perceptual loss computes feature-level differences using a pre-trained deep network, encouraging semantic consistency between reconstructed and ground truth images. Edge-aware and Dice-based losses focus on boundary regions, promoting sharper edges and finer structural details. Total variation loss is introduced as a regularization term to suppress high-frequency artifacts and improve spatial smoothness.

The final training objective is defined as a weighted combination of the above loss terms:

$$\mathcal{L}_{\mathrm{total}} = \lambda_{\mathrm{L1}}\mathcal{L}_{\mathrm{L1}} + \lambda_{\mathrm{SSIM}}\mathcal{L}_{\mathrm{SSIM}} + \sum_k \lambda_k \mathcal{L}_k, \tag{10}$$

where λ denotes the weighting coefficient for each loss component.

3 Experiments and Results

In this section, we evaluate the effectiveness of deep neural networks for recovering visually meaningful plaintext images from encrypted inputs. A series of controlled experiments are conducted to analyze the impact of network architecture and loss function design on decryption quality.

Specifically, we compare multiple configurations, including a standard U-Net, an attention-enhanced U-Net, and different loss formulations, in order to investigate how network structure and selection of loss function influence reconstruction performance. Both quantitative loss trends and qualitative visual results are presented to provide a comprehensive assessment.

3.1 Dataset and Experimental Setup

All experiments are conducted on the Caltech-101 dataset[6], a widely used benchmark containing natural images from 101 object categories and an additional background class. Each category includes approximately 40 to 800 images, with most classes containing around 50 samples, resulting in a total of about 9,000 images.

To adapt the dataset for the DRPE decryption task, all images are converted to grayscale and resized to a fixed spatial resolution of 128×128 pixels. Prior to encryption, pixel intensities are normalized to ensure numerical stability during network training. Each plaintext image is then encrypted using the double random phase encryption (DRPE) scheme described in Section 2.1, generating the corresponding encrypted amplitude images as network inputs.

For each experimental run, the dataset is divided into a training set (80% of images per category) and a test set (20% per category).This split is independently performed to reduce potential bias and improve the robustness of evaluation. Unless otherwise specified, the same data split is used across different model configurations to ensure fair comparison.

All neural networks are trained to learn a direct mapping from encrypted images to their corresponding plaintext images. Training and evaluation are performed on the same resolution without multi-scale augmentation, allowing the influence of network architecture and loss function design to be isolated and analyzed.

The overall experimental pipeline, including DRPE encryption, neural network training, and decryption evaluation, is illustrated in Fig. 3.

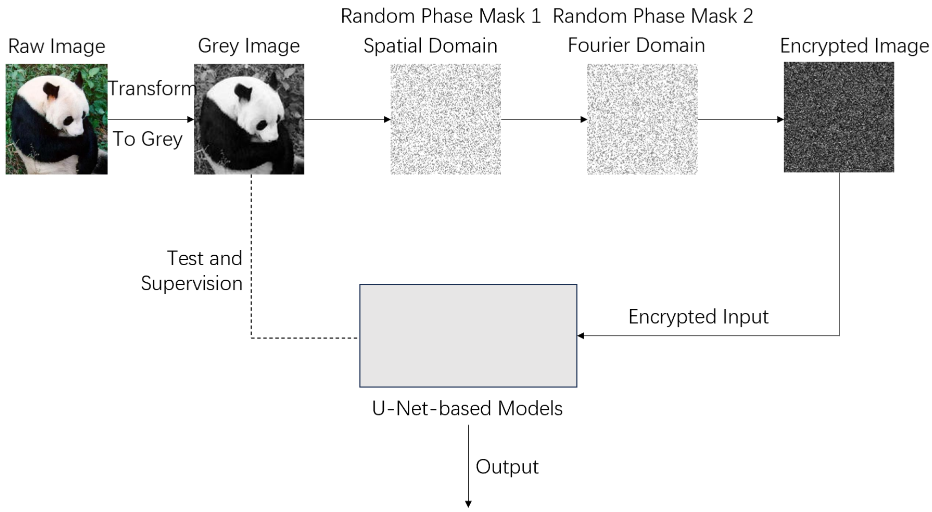

Figure 3: Overall experimental pipeline. Plaintext images are first encrypted using the double random phase encryption (DRPE) scheme. The encrypted amplitude images are then fed into a U-Net-based neural network to learn the inverse mapping and recover the original plaintext images.

3.2 Comparison of Loss Functions and Training Strategies

To systematically analyze the impact of different loss functions on DRPE image reconstruction, we evaluate five representative training configurations, including L1, L1 + Attention, SSIM + L1, SSIM + L1 + Attention, and multi-loss refinement strategies. Both quantitative convergence behavior and qualitative reconstruction quality are examined.

3.2.1 Loss Convergence Analysis

Figure 4 presents the training loss curves for each loss configuration over the entire training process. For clarity, each loss function is plotted separately to highlight its convergence characteristics. As shown in Fig. 4, all configurations demonstrate stable convergence without divergence or oscillation, indicating that the proposed network architectures are trainable under various loss formulations and have few room for optimization. However, differences in convergence speed and final loss values can be observed.

The L1-based models converge faster and reach lower steady-state loss values compared to SSIM-related losses. In particular, the L1 + Attention configuration achieves the lowest final loss, suggesting that pixel-wise supervision combined with adaptive feature weighting provides the most effective optimization signal for DRPE reconstruction.

3.2.2 Joint Comparison of Loss Functions

To facilitate a direct comparison, Fig. 5 overlays the loss curves of all five configurations within a single plot. From Fig. 5, it is evident that loss functions incorporating SSIM exhibit higher final losses and slower convergence. This behavior is consistent with the observation that perceptual similarity metrics impose additional structural constraints that may conflict with the heavily

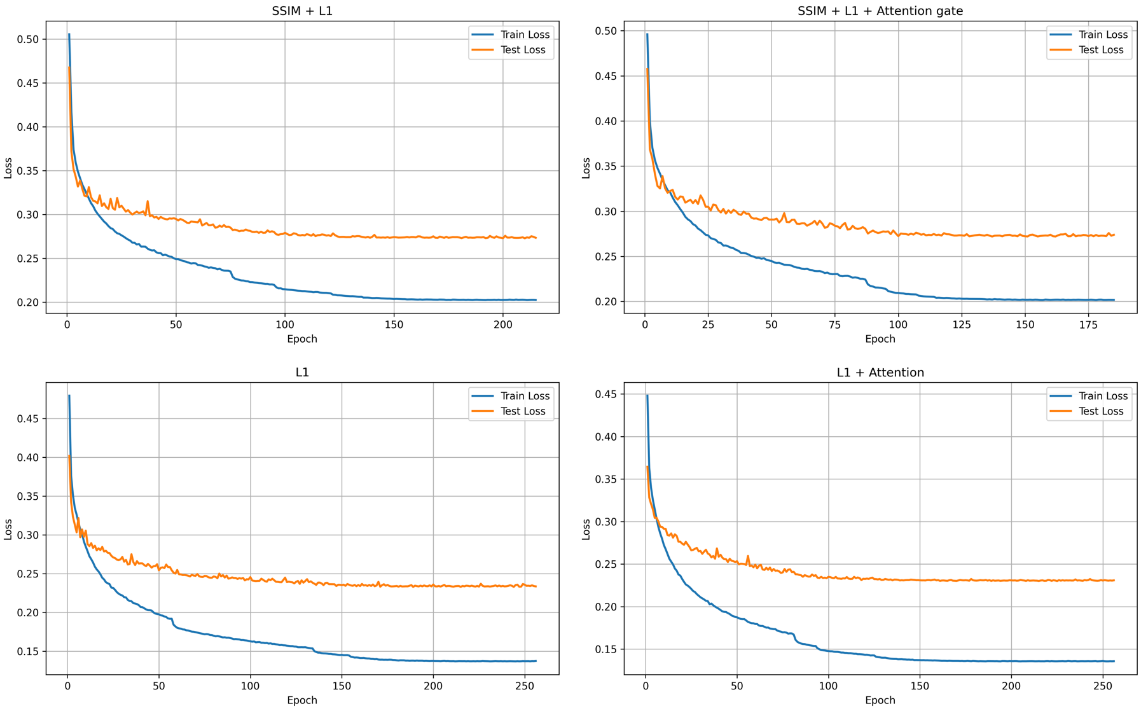

Figure 4: Training loss curves under different loss functions plotted separately for clarity.

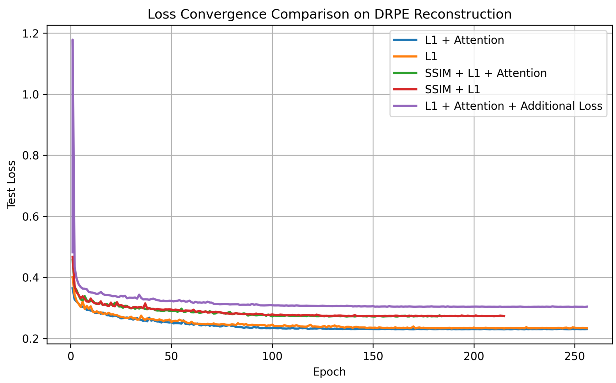

Figure 5: Joint comparison of training loss curves for all five loss configurations.

distorted nature of DRPE-encrypted inputs.

Although multi-loss strategies shows similar convergence speed as L1-based losses, the improvement remains marginal. These results suggest that the reconstruction performance may be primarily limited by information availability rather than optimization strategy alone.

3.3 Visual Evaluation and Recoverability Analysis

Figure 6 presents the decrypted results of different models on a soccer image, which serves as a representative example with clear geometric patterns and moderate structural complexity.

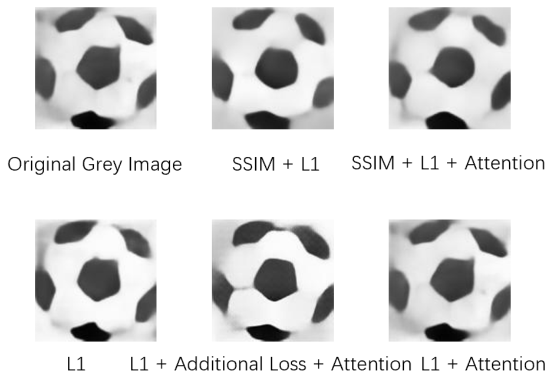

Figure 6: Visual comparison of decrypted soccer images produced by different models.

As shown in Fig. 6, all evaluated models are able to successfully recover the overall shape of the soccer, demonstrating that learning-based approaches can effectively invert DRPE-encrypted magnitude-only inputs. However, noticeable differences arise when inspecting local geometric details. In particular, models incorporating SSIM loss tend to smooth sharp intensity transitions, causing the polygonal black leather patches on the soccer surface to appear rounded and less angular. In contrast, the multi-loss configuration dominated by L1 supervision preserves polygon edges more faithfully, yielding the clearest representation of the original geometric boundaries.

It is worth noting that models trained with pure L1 loss and L1 combined with attention preserve an additional angular corner in certain black polygon regions, which is also present in the original image. This observation highlights the advantage of pixel-wise supervision in recovering sharp intensity transitions and subtle geometric details. However, such sensitivity to local pixel variations may also lead to slight structural exaggeration in regions with dense or repetitive high-frequency patterns.

To further differentiate the reconstruction behaviors of different loss functions and network designs, Fig. 7 shows a second group of decrypted examples based on a panda image, where visual discrepancies among models become more pronounced.

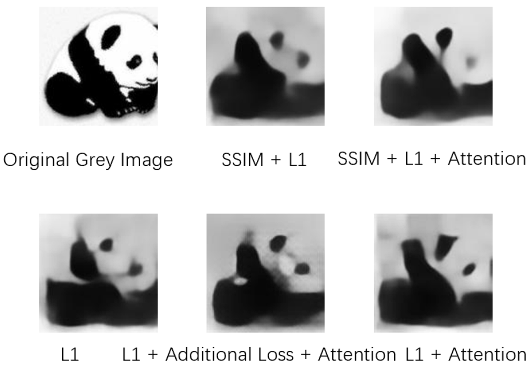

Figure 7: Visual comparison of decrypted panda images highlighting differences in detail reconstruction.

As illustrated in Fig. 7, attention-enhanced models produce relatively sharper contours and more coherent fine structures, particularly around the facial region. Among all configurations, the L1 + Attention model achieves the most visually plausible reconstruction by correctly recovering the panda's right eye, which is either blurred or misplaced in other models. In contrast, reconstructions obtained using pure L1 loss and L1-based multi-loss refinement fail to preserve the characteristic appearance of the panda, resulting in images that deviate significantly from the expected semantic structure.

Despite these improvements, none of the evaluated models are able to recover certain fine details, such as the panda's mouth and the upper-right ear region. This consistent failure across models suggests the presence of intrinsic limitations in magnitude-only DRPE inversion, which will be further discussed in the following subsection on model limitations.

3.4 Limitations

Despite its overall effectiveness, the proposed L1+Attention model still exhibits limitations when dealing with images containing highly complex textures and dense high-frequency details. As illustrated in Fig. 8, though the main structure of sunflowers is decrypted, fine structures such as the petal boundaries of the sunflower image cannot be fully recovered, resulting in blurring and loss of local detail. This suggests that magnitude-only DRPE encryption inherently removes critical phase information, imposing an upper bound on reconstruction fidelity even for the best-performing model.

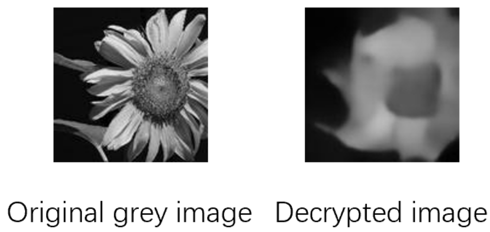

Figure 8: Limitation of the proposed model

4 Discussion

This work investigates the effectiveness of deep learning–based reconstruction for Double Random Phase Encoding (DRPE) encrypted images under various network architectures and loss configurations. Several important observations can be drawn from the experimental results.

First, though perceptual similarity losses such as SSIM are widely adopted in natural image restoration tasks, our experiments show that combining SSIM with L1 loss does not outperform the use of L1 loss alone in the DRPE reconstruction task. This phenomenon can be attributed to the fundamental characteristics of DRPE encryption. Since DRPE severely disrupts local spatial structures through phase modulation and Fourier-domain transformations, enforcing perceptual consistency may introduce optimization conflicts. In contrast, pixel-wise L1 loss provides a more direct and stable supervision signal for recovering amplitude information from encrypted observations.

Second, the introduction of attention mechanisms leads to noticeable improvements in reconstruction quality. Attention modules enable the network to adaptively emphasize informative regions and suppress noise-like responses introduced by the encryption process. This is particularly beneficial for DRPE decryption, where recoverable information is unevenly distributed across spatial and frequency domains. The superior performance of the L1 + Attention configuration suggests that selectively enhancing salient features is more effective than enforcing global perceptual constraints.

Third, we observe that further refinement using multiple loss functions, which has been carefully tuned, yields only marginal performance improvement. As shown in Fig. 9, while adding losses concentrating on detail recovery to refine the best model, the test loss remain stable and has almost no signs of decline. This behavior indicates the presence of an intrinsic performance ceiling. One persuasive explanation is that the use of magnitude-only encrypted images inevitably discards phase information, resulting in an irreversible loss of certain image details. As a result, no learning-based method can fully recover information that is fundamentally absent from the input.

Despite some encouraging results, several limitations remain. The weighting of multiple loss functions relies on empirical tuning, which limit the performance of best results. In addition, the current model does not explicitly incorporate the uncertainty or variability of DRPE keys, which may limit its robustness under unknown or dynamically changing encryption parameters. Concurrently, the reconstruction quality for extremely small or low-contrast target regions remains suboptimal, indicating potential underfitting in highly localized areas. Finally, the generalization capability of the proposed network under varying DRPE system parameters, such as random phase distributions and optical configurations, remains uncertain.

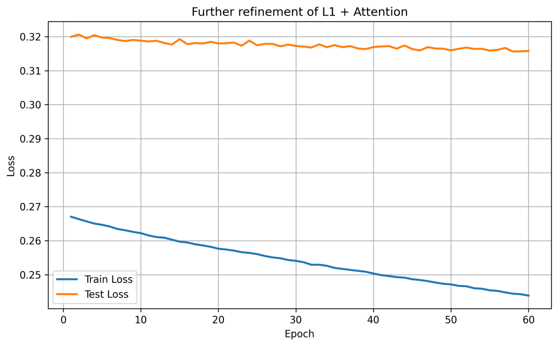

Figure 9: Loss of further refinement of L1+Attention model

5 Conclusion

In this paper, we demonstrate that deep learning–based approaches can effectively reconstruct images encrypted by traditional Double Random Phase Encoding schemes, even when only magnitude information is available. Through extensive experiments, we show that convolutional architectures enhanced with attention mechanisms significantly improve reconstruction quality by selectively emphasizing recoverable information.

Our results further reveal that, contrary to common practice in natural image restoration, perceptual similarity losses such as SSIM provide limited benefits in the DRPE reconstruction task. Instead, pixel-wise objectives combined with attention mechanisms yield more stable optimization and superior performance. While multi-loss refinement offers marginal improvements, the overall reconstruction quality appears bounded by information loss inherent to the encryption process.

These findings suggest that the security of classical optical encryption systems should be carefully re-evaluated in the era of data-driven learning. The ability of neural networks to approximate inverse mappings without explicit knowledge of encryption keys highlights potential vulnerabilities in DRPE-based schemes.

References

[1] P, Refregier, B, Javidi. Optical image encryption based on input plane and Fourier plane random encoding.[J]. Optics letters,1995,20767-9.

[2] Olaf Ronneberger, Philipp Fischer, Thomas Brox. U-Net: Convolutional Networks for Biomedical Image Segmentation[C]//Medical image computing and computer-assisted intervention – MICCAI 2015 : Part III /.:Springer,, 2015:234-241.

[3] ZHAO Yangniu, ZHONG Guoqiang, YU Hui. A review on the attention mechanism of deep learning[J]. Neurocomputing, 2021, 452: 48-62.

[4] Oktay O, Schlemper J, Folgoc L L, et al. Attention U-Net: Learning Where to Look for the Pancreas[EB/OL]. (2018-04-11)[2025-12-14].

[5] NILSSON J, AKENINE-MÖLLER T. Understanding SSIM[EB/OL]. (2020)[2025-12-21].

[6] Li Fei-Fei, Rob Fergus, and Pietro Perona. One-shot learning of object categories. IEEE Trans. Pattern Anal. Mach. Intell., 28(4):594–611, April 2006.